Homework#6: Pre-process the Adult Dataset (for Predicting Socio-Economic Status)

Objective

Perform Exploratory Data Analysis (EDA) and Pre-processing on the **Adult Income Dataset**. Your goal is to transform raw, messy data into a clean, numerical format ready for predicting if **income** is >50K or <=50K) in the next step

Phase 1: Exploratory Data Analysis (EDA)

Goal: Understand the "shape" and "health" of your data.

- **Descriptive Statistics:** Generate a table of summary statistics (mean, median, mode, std dev, skewness) for **numerical features.**
  - **Interpretation:** Identify which numerical features have high variance or significant skew.
- **Data Quality Audit:**
  - Identify the data type (Categorical vs. Numerical) for every column.
  - Count missing values. *Note: In this dataset, missing values are often coded as a character (e.g., '?'). Find and report these.*
- **Categorical Analysis:**
  - Identify every distinct "label" or "category" present in each of categorical columns.
  - Compare education and education-num. Explain why keeping both might be redundant.

Phase 2: Visual Discovery & Outlier Handling

Goal: Use visuals to justify your cleaning strategy.

- **Visualizing Distributions:** Create histograms for age, hours-per-week, and capital-gain.
  - Create box plots for numerical variables to detect outliers.
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- **Outlier Strategy:** Pick one feature with extreme values (e.g., capital-gain or age).
  - **Decision:** State whether you will **remove**, **replace**, or **keep** these outliers. Justify your choice based on whether you think they are "errors" or "real-world extremes.

Phase 3: Data Pre-processing & Transformation

Goal: Prepare the data for a machine learning algorithm.

- **Handling Missing Values:** Apply an imputation strategy. Explain why you chose your specific method.
- **Categorical Encoding:** Convert text to numbers. Use **One-Hot Encoding** for nominal data (e.g., occupation) and **Label/Ordinal Encoding** for data with a clear rank (e.g., education).

Phase 4: Scaling & Final Preparation

Goal: Normalize the data range.

- **Feature Scaling:** Apply **Standardization** or **Min-Max Scaling** to your numerical features.

Submission Requirements

- **Format:** A Jupyter Notebook (.ipynb) or a RapidMiner package (.xml + .pdf report).